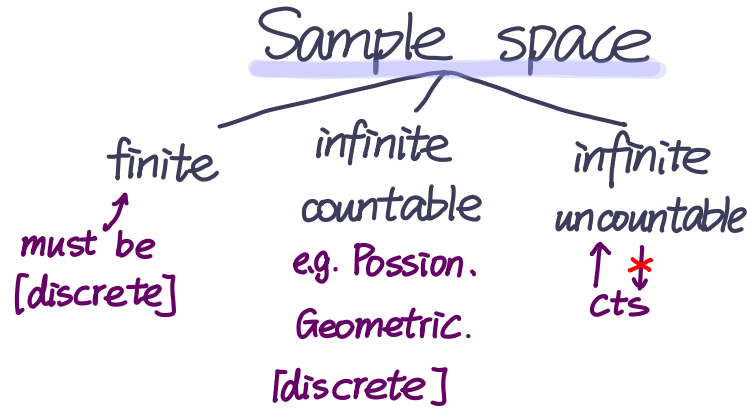
STA2002: Probability and Statistics II
Confidence Interval

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Discrete v.s. Cts.



Bernoulli p. $p \in [0,1]$.

Binomial (n,p) $p \in [0,1]$ $n \in N^+$

$X \in \{0,1,2 \dots n\}$ (sample space)

$P(x) = C_n^x p^x (1-p)^{n-x}$

Introduction

- In this lecture we would introduce the technique of confidence interval (CI), which is an important way to construct interval estimator for a parameter.
- Confidence intervals for means and confidence intervals for the difference of two means would be discussed.
- **Suggested reading**: Chapter 7.1 and 7.2 of the textbook.

Confidence Interval

Definition 1 (Confidence Interval)

Let $X_1, X_2 \ldots, X_n$ be a sample on a random variable X, where X has pdf $f(x;\theta)$, $\theta \in \Omega$. Let $0 < \alpha < 1$ be specified. Let $L = L(X_1, X_2, \ldots, X_n)$ and $U = U(X_1, X_2, \ldots, X_n)$ be two statistics. We say that the interval (L, U) is a $(1-\alpha)100\%$ ***confidence interval*** *for θ is*

a function of sample. ↖ random

Random.

$$1 - \alpha = P_\theta[\theta \in (L, U)].$$

true value (unknown) A constant. [Not Random]

That is, the probability that the interval includes θ is $1-\alpha$, which is called the ***confidence coefficient*** *of the interval.*

(Two-sided) Confidence Intervals for μ

Suppose that $X_1, \ldots, X_n$, i.i.d., is a random sample of size n. We are interested in estimating the mean $\mu := E(X_1)$. We will discuss 4 cases:

- Case 1: $X_1, \ldots, X_n \overset{i.i.d.}{\sim} N(\mu, \sigma^2)$ and σ^2 is known
- Case 2: $X_1, \ldots, X_n$ are i.i.d. with mean $\mu = E(X_1)$ and variance $\sigma^2 = \mathrm{Var}(X_1)$, and σ^2 is known
- Case 3: $X_1, \ldots, X_n \overset{i.i.d.}{\sim} N(\mu, \sigma^2)$ and σ^2 is unknown
- Case 4: $X_1, \ldots, X_n$ are i.i.d. with mean $\mu = E(X_1)$ and variance $\sigma^2 = \mathrm{Var}(X_1)$, and σ^2 is unknown

Key idea: first construct a statistic involving μ.
the distribution of the statistic can be drived.

(Two-sided) CI for μ: Case 1

- Let

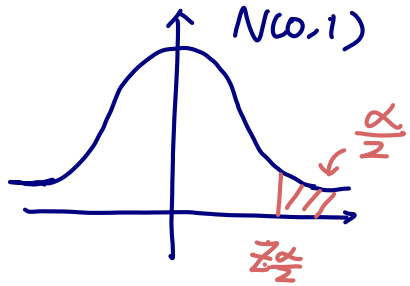
$$Z = \underbrace{\frac{\overline{X} - \mu}{\sigma/\sqrt{n}}}_{\text{statistic}} \sim N(0,1).$$

sample mean

- Please show and confirm that, the probability that the random interval

$$\left[\overline{X} - z_{\alpha/2}\left(\frac{\sigma}{\sqrt{n}}\right), \overline{X} + z_{\alpha/2}\left(\frac{\sigma}{\sqrt{n}}\right)\right]$$

includes μ is $1 - \alpha$.



(Two-sided) CI for μ: Case 1

- Once the sample is observed, we can compute the sample mean $\overline{x}$. The computed interval

$$\overline{x} \pm z_{\alpha/2}(\sigma/\sqrt{n})$$

  is called a $100(1-\alpha)\%$ **confidence interval (C.I.)** of μ.

- Note that this constructed C.I. of μ has the following properties,
  - The C.I. is centered at $\overline{x}$, a point estimate of μ, and the C.I. has a width of $2z_{\alpha/2}(\sigma/\sqrt{n})$.
  - The bigger the sample size n, the smaller the width of the C.I., resulting in a shorter C.I.
  - The bigger the α, the smaller the z_α, resulting in a shorter C.I.

for a fixed α, can have different C.I. $\Rightarrow$ smaller is better

(Two-sided) CI for μ: Case 1

- Here we show some deduction details. Recall that $Z = \dfrac{\overline{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$. Then,

$$P\left(-z_{\alpha/2} \leq \frac{\overline{X} - \mu}{\sigma/\sqrt{n}} \leq z_{\alpha/2}\right) = 1 - \alpha$$

$$P\left(-z_{\alpha/2}\left(\frac{\sigma}{\sqrt{n}}\right) \leq \overline{X} - \mu \leq z_{\alpha/2}\left(\frac{\sigma}{\sqrt{n}}\right)\right) = 1 - \alpha$$

$$P\left(-\overline{X} - z_{\alpha/2}\left(\frac{\sigma}{\sqrt{n}}\right) \leq -\mu \leq -\overline{X} + z_{\alpha/2}\left(\frac{\sigma}{\sqrt{n}}\right)\right) = 1 - \alpha$$

$$P\left(\overline{X} + z_{\alpha/2}\left(\frac{\sigma}{\sqrt{n}}\right) \geq \mu \geq \overline{X} - z_{\alpha/2}\left(\frac{\sigma}{\sqrt{n}}\right)\right) = 1 - \alpha.$$

(Two-sided) CI for μ: Case 1

Example 2

This is Example 7.1-1 in book. Let X be the length of life of a light bulb, and assume that $X \sim N(\mu, 1296)$. We have a random sample of $n = 27$ with $\overline{x} = 1478$. A 95% confidence interval for μ is

$$\left[\overline{x} - z_{0.025}\left(\frac{\sigma}{\sqrt{n}}\right), \overline{x} + z_{0.025}\left(\frac{\sigma}{\sqrt{n}}\right)\right]$$

$$= \left[1478 - 1.96\left(\frac{36}{\sqrt{27}}\right), 1478 + 1.96\left(\frac{36}{\sqrt{27}}\right)\right]$$

$$= [1464.42, 1491.58].$$

(Two-sided) CI for μ: Case 2

- Case 2: $X_1, \ldots, X_n$ are i.i.d. with mean $\mu = E(X_1)$ and variance $\sigma^2 = \mathrm{Var}(X_1)$, and σ^2 is known.
- Note that $X_1, \ldots, X_n$ do not necessarily follow a normal distribution.
- When n is large enough, by central limit theorem,

$$\frac{\overline{X} - \mu}{\sigma/\sqrt{n}} \overset{\text{approx}}{\sim} N(0,1).$$

recall "CLT" (review: proof by using mgf)

require i.i.d.

NOT require distribution

(Two-sided) CI for μ: Case 2

- As a result, approximately we have

$$P\left(-z_{\alpha/2} \leq \frac{\overline{X}-\mu}{\sigma/\sqrt{n}} \leq z_{\alpha/2}\right) \approx 1-\alpha$$

$$P\left(\overline{X} - z_{\alpha/2}\left(\frac{\sigma}{\sqrt{n}}\right) \leq \mu \leq \overline{X} + z_{\alpha/2}\left(\frac{\sigma}{\sqrt{n}}\right)\right) \approx 1-\alpha.$$

- $\overline{x} \pm z_{\alpha/2}(\sigma/\sqrt{n})$ is called an **approximate** $100(1-\alpha)\%$ **confidence interval (C.I.)** of μ.

(Two-sided) CI for μ: Case 2

Example 3

This is Example 7.1-3 in book. Let X be the amount of orange juice consumed by an American per day. We know that $\sigma = 96$. To estimate μ, we have a sample of $n = 576$ and $\overline{x} = 133$. An approximate 90% confidence interval for μ is

$$133 \pm 1.645\left(\frac{96}{\sqrt{576}}\right) = [126.42, 139.58].$$

(Two-sided) CI for μ: Case 3

- Case 3: $X_1, \ldots, X_n \overset{i.i.d.}{\sim} N(\mu, \sigma^2)$ and σ^2 is unknown
- As σ is unknown, we can not use the C.I. in Case 1 since the interval

$$\overline{x} \pm z_{\alpha/2}(\sigma/\sqrt{n})$$

depends on σ.

- Instead, recall the sample variance

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2,$$

and

$$T = \frac{\overline{X} - \mu}{S/\sqrt{n}} \sim t(n-1).$$

shifting & transform
identical : can be achieved by
independent : strong assumption

Theorem 11 (Student's Theorem)

Let $X_1, \ldots, X_n$ be i.i.d. random variables each having a normal distribution with mean μ and variance σ^2. Define the random variables

$$\overline{X} := \frac{1}{n}\sum_{i=1}^{n} X_i, \quad \text{and} \quad S^2 := \frac{1}{n-1}\sum_{i=1}^{n}(X_i - X)^2.$$

Then

(a) $\overline{X}$ has a $N(\mu, \frac{\sigma^2}{n})$ distribution.

(b) $\overline{X}$ and S^2 are independent.

(c) $(n-1)S^2/\sigma^2$ has a $\chi^2(n-1)$ distribution.

(d) The random variable

$$T = \frac{\overline{X} - \mu}{S/\sqrt{n}}$$

has a Student t-distribution with $n-1$ degrees of freedom.

$S_n^2 \longrightarrow \sigma^2$ as $n \to \infty$

student's thm

$t(n) \xrightarrow{D} N(0,1)$ as $n \to \infty$

(Two-sided) CI for μ: Case 3

- Consider

$$P\left(-t_{\alpha/2}(n-1) \leq \frac{\overline{X}-\mu}{S/\sqrt{n}} \leq t_{\alpha/2}(n-1)\right) = 1-\alpha$$

$$P\left(\overline{X} - t_{\alpha/2}(n-1)\left(\frac{S}{\sqrt{n}}\right) \leq \mu \leq \overline{X} + t_{\alpha/2}(n-1)\left(\frac{S}{\sqrt{n}}\right)\right) = 1-\alpha.$$

- Once we have the realized sample, we can compute $\overline{x}$ and s, and

$$\left[\overline{x} - t_{\alpha/2}(n-1)\left(\frac{s}{\sqrt{n}}\right), \overline{x} + t_{\alpha/2}(n-1)\left(\frac{s}{\sqrt{n}}\right)\right]$$

is called a $100(1-\alpha)\%$ **confidence interval (C.I.)** of μ.

(Two-sided) CI for μ: Case 3

Example 4

This is Example 7.1-5 in book. Let $X \sim N(\mu, \sigma^2)$, and both μ and σ^2 are unknown. We compute that $n = 20$, $\overline{x} = 507.50$ and $s = 89.75$. Since $t_{0.05}(19) = 1.729$, a 90% confidence interval for μ is

$$507.50 \pm 1.729\left(\frac{89.75}{\sqrt{20}}\right) = [472.80, 542.20].$$

(Two-sided) CI for μ: Case 4

- Case 4: $X_1, \ldots, X_n$ are i.i.d. with mean $\mu = E(X_1)$ and variance $\sigma^2 = \mathrm{Var}(X_1)$, and σ^2 is unknown.
- If n is large enough (say $n \geq 100$), or if each X_i is approximately normal, then

$$\frac{\overline{X} - \mu}{S/\sqrt{n}} \overset{\text{approx}}{\sim} t(n-1).$$

- As a result,

$$\left[\overline{x} - t_{\alpha/2}(n-1)\left(\frac{s}{\sqrt{n}}\right), \overline{x} + t_{\alpha/2}(n-1)\left(\frac{s}{\sqrt{n}}\right)\right]$$

is called an **approximate** $100(1-\alpha)\%$ **confidence interval (C.I.)** of μ.

(Two-sided) CI for μ: Case 4

Note that,

- If n is large enough, then $t_{\alpha/2}(n-1) \approx z_{\alpha/2}$, and so we can also use the interval

$$\left[\overline{x} - z_{\alpha/2}\left(\frac{s}{\sqrt{n}}\right), \overline{x} + z_{\alpha/2}\left(\frac{s}{\sqrt{n}}\right)\right]$$

  as an approximate $100(1-\alpha)\%$ C.I. for μ.
- If n is small or when each X_i does not look like a normal distribution (say when X_i are very skewed), this method has to be used with caution.

One-sided C.I. for μ

- Up until now we have only been discussing **two-sided** confidence intervals, since the C.I. are often of the form $\overline{x} \pm a$ for some constants a that depends on α, n and the samples.
- Suppose that $X_1, \ldots, X_n \overset{i.i.d.}{\sim} N(\mu, \sigma^2)$ and σ^2 is known. Consider

$$P\left(\frac{\overline{X} - \mu}{\sigma/\sqrt{n}} \leq z_\alpha\right) = 1 - \alpha,$$

$$P\left(\overline{X} - z_\alpha\left(\frac{\sigma}{\sqrt{n}}\right) \leq \mu\right) = 1 - \alpha.$$

One-sided C.I. for μ

- As a result, we have

$$\left[\overline{x} - z_\alpha\left(\frac{\sigma}{\sqrt{n}}\right), \infty\right)$$

an **one-sided** $100(1-\alpha)\%$ **confidence interval** for μ.

- Similarly, we can show that

$$\left(-\infty, \overline{x} + z_\alpha\left(\frac{\sigma}{\sqrt{n}}\right)\right]$$

is another one-sided $100(1-\alpha)\%$ confidence interval for μ.

C.I. for the Difference of Two Means

- Suppose that we have two populations from which we draw i.i.d. random samples, say $X_1, \ldots, X_n \sim N(\mu_X, \sigma_X^2)$ and $Y_1, \ldots, Y_m \sim N(\mu_Y, \sigma_Y^2)$.
- We would like to construct confidence intervals for $\mu_X - \mu_Y$, the difference between the two population means.
- We will the following case in details,

  $X_1, \ldots, X_n$ and $Y_1, \ldots, Y_m$ are independent. And $\sigma_X^2 = \sigma_Y^2 = \sigma^2$, and σ^2 is unknown (Two populations have the same variance σ^2)

Two-sample Pooled t-interval

Theorem 5

Let $X_1, \ldots, X_n \overset{i.i.d.}{\sim} N(\mu_X, \sigma^2)$ *and* $Y_1, \ldots, Y_m \overset{i.i.d.}{\sim} N(\mu_Y, \sigma^2)$ *be independent random variables. Then a* $100(1-\alpha)\%$ *C.I. for* $\mu_X - \mu_Y$ *is*

$$\overline{X} - \overline{Y} \pm t_{\alpha/2}(n+m-2) S_p \sqrt{\frac{1}{n} + \frac{1}{m}},$$

where S_p^2 *is the pooled estimator of* σ^2*:*

$$S_p^2 = \frac{(n-1)S_X^2 + (m-1)S_Y^2}{n+m-2} = \frac{\sum_{i=1}^n (X_i - \overline{X})^2 + \sum_{i=1}^m (Y_i - \overline{Y})^2}{n+m-2},$$

which is an unbiased estimator of σ^2*.*

Proof. The proof would be discussed in class. □

Two-sample Pooled t-interval

Example 6

This is Example 7.2-2 in book. Let $X \sim N(\mu_X, \sigma^2)$ be the score on a standardized test in a large high school, and $Y \sim N(\mu_Y, \sigma^2)$ be the score on a standardized test in a small high school. We have

$$
\begin{aligned}
n &= 9 \\
\overline{x} &= 81.31 \\
s_x^2 &= 60.76 \\
m &= 15 \\
\overline{y} &= 78.61 \\
s_y^2 &= 48.24.
\end{aligned}
$$

Two-sample Pooled t-interval

Solution. To construct a 95% confidence interval for $\mu_X - \mu_Y$, we calculate

$$t_{0.025}(22) = 2.074$$

$$s_p = \sqrt{\frac{8(60.76) + 14(48.24)}{22}}$$

The required 95% confidence interval for $\mu_X - \mu_Y$ is

$$81.31 - 78.61 \pm 2.074 \sqrt{\frac{8(60.76) + 14(48.24)}{22}} \sqrt{\frac{1}{9} + \frac{1}{15}} = [-3.65, 9.05].$$

□